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Effect of school climate on fourth-grade students' science achievement in the United Arab Emirates: Insights from the 2023 TIMSS Dataset

Hassan Tairab [a], Latifa Alketbi [a], Ahmad Qablan [a], Laurent Gabriel Ndijuye [a], Maxwell Peprah Opoku [b,*]

[a] Curriculum and Methods of Instruction, College of Education, United Arab Emirates University, Al Ain, , United Arab Emirates
[b] Special and Gifted Education Department, College of Education, United Arab Emirates University, Al Ain, United Arab Emirates

A R T I C L E I N F O

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A B S T R A C T

Studies have documented the relationship between school climate and science achievement. However, discussions on school climate are very general as limited attempts have been made to explore the form of school climate which could impact on student achievement in science. Guided by the Systems View of School Climate conceptual framework and using the 2023 dataset of the Trends in International Mathematics and Science Study (TIMSS), we explored the relationship between school climate and science achievement from the perspective of students in the United Arab Emirates (UAE). The study population comprised 34,217 fourth-grade students who participated in the TIMSS test, including 16,784 females (49 %) and 17,433 males (51 %). Pearson's correlation coefficient, hierarchical multiple regressions and moderation analyses were employed to identify predictors of science achievement. The three tenets of school climate made substantial contribution in the variance in science achievement. Given the UAE's recent substantial investments in science education, these findings reflect the importance of fostering a supportive school climate to enhance students' science achievement.

Introduction

In recent years, high-income countries have been participating in international assessments to evaluate the efficiency and effectiveness of their education systems. One of the most prominent international assessments is the Trends in International Mathematics and Science Study (TIMSS; Wardat et al., 2022; Zhang & Bae, 2020). For over three decades, TIMSS has been administered by the International Association for the Evaluation of Educational Achievement (IEA) in >56 countries. Since 2011, the United Arab Emirates (UAE) has participated in TIMSS assessments for fourth and eighth grades (Ministry of Education [MoE], 2021). Using the TIMSS 2019 dataset, this study examines how school climate, which is an aspect of learning environment, impacts the science achievements of fourth-grade students in the UAE. By identifying key predictors, the research aims to contribute to curriculum reforms and improve science learning outcomes.

Learning environment encompasses social, cultural, psychological and physical settings where learning takes place (Rusticus et al., 2020, 2023). Learning takes place within an environment which is made up of people from diverse backgrounds (Lorsbach & Jinks, 1999). For instance, within a school environment, there are ongoing interactions between stakeholders such as teachers, students, allied professionals and even external agents (Lorsbach & Jinks, 1999). This means that different motivations, emotions and experiences converge at each stage of acquisition of knowledge among students (Rusticus et al., 2020, 2023). Consequently, the physical and social settings of a learning environment have influence on the success of students (Lorsbach & Jinks, 1999). Students' perception of the learning environment is vital to unearthing factors which could have impact on the learning experiences of students (Lorsbach & Jinks, 1999). In this study, attempts were made to understand the relationship between students' achievement in science and school climate which is used as a proxy for learning environment.

Theoretical frameworks such as Vygotsky's theory and the bioecological model, along with models developed by Epstein, Hoover–Dempsey, and Sandler, explain the relationship between the learning environment and student achievement (Demirtas-Zorbaz, Akin-Arikan & Terzi, 2021; Teng, 2020). Although these theories and models provide insights into the role of school climate, less is known about risk factors, such as how disadvantaged family backgrounds and poor school climate negatively impact learning outcomes. Conversely,

* Correspondence author at. Special Education Department, College of Education, United Arab Emirates University, P.O. Box 15551, Al-Ain, United Arab Emirates.
E-mail address: maxwell.p@uaeu.ac.ae (M.P. Opoku).

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empirical evidence consistently indicates that protective factors strengthen students' resilience by compensating for or moderating the adverse effects of risk factors (Gómez & Suárez, 2020; Guzey & Li, 2023; Zysberg & Schwabsky, 2021). Evidence implies that although a disadvantaged family background may serve as a risk factor, a positive school climate may serve as a protective factor that plays both a compensatory and moderating role (Gómez & Suárez, 2020; Guzey & Li, 2023).

Studies conducted across various contexts have documented the critical role of school climate in students' academic and non-academic achievements (Gómez & Suárez, 2020; Guzey & Li, 2023; Teng, 2020; Winberg & Palm, 2021). The importance of school climate in promoting school effectiveness has garnered increasing attention from researchers, policymakers, parents, and other stakeholders (Guzey & Li, 2023; Teng, 2020). In countries such as the United States (Beller, 2013), the United Kingdom (Zysberg & Schwabsky, 2021), and more recently, China (Teng, 2020), educational policymakers have prioritized fostering a positive school climate, launching several governmental initiatives to support this goal. Internationally, empirical evidence consistently indicates that a positive school climate, which is characterized by disciplined environmental factors, strong student–teacher relationships, and high teacher expectations, significantly improves students' academic and nonacademic performance (Berkowitz et al., 2015; Teng, 2020; Winberg & Palm, 2021).

Some studies have linked school climate with educational equality and equity (Guzey & Li, 2023; Teng, 2020). Research suggests that a positive school climate significantly enhances students' academic achievements even after accounting for background variables (Kutsyuruba et al., 2015; Teng, 2020). It is argued that school climate plays a compensatory role in bridging gaps in academic achievement related to students' family backgrounds. The rationale is that if school climate is particularly influential in schools facing adversity, then improving school climate in these schools could help narrow academic achievement gaps (Teng, 2020).

There is consensus among educational researchers and stakeholders that a positive school climate fosters student development and enhances learning outcomes (Gómez & Suárez, 2020; Guzey & Li, 2023); however, this topic has not been empirically studied in the context of the UAE. This gap is even more evident regarding the specific effects of school climate on science teaching and learning. Given the wealth of data generated by the UAE's participation in TIMSS, this study provides an opportunity to examine how school climate influences students' science achievement. It seeks to contribute to the literature by analyzing the role of school climate in shaping UAE students' science achievement in the TIMSS 2019 assessment.

Theoretical framework

In this study, school climate is conceptualized as students' perceptions regarding interactions with peers, their sense of safety, and their beliefs about learning (Rudasill et al., 2018). A positive school climate suggests a favorable disposition among students toward several affective constructs (Wang & Degol, 2016). Due to this complexity, one of the main theoretical lenses consistently used in studies on school climate is Bronfenbrenner's ecological systems theory (EST), which helps explain the complex factors affecting child development (Bronfenbrenner, 1989, 1992, 1994). Within the school context, students interact in a complex environment that involves interactions with multiple actors (Rudasill et al., 2018). The EST comprises five components: the microsystem, mesosystem, exosystem, macrosystem, and chronosystem (Bronfenbrenner, 1989, 1992, 1994). The microsystem includes immediate environments such as school and family, where individuals engage in direct interactions (Bronfenbrenner, 1989, 1992, 1994). The next level, the mesosystem, describes interactions within the microsystem (Bronfenbrenner, 1989, 1994). The exosystem and macrosystem represent broader contextual influences that, although not directly experienced by the child, still affect their development (Bronfenbrenner, 1992). For instance, the exosystem includes connections between settings in which a child is not directly involved but that still impact their development, such as a parent's workplace (Bronfenbrenner, 1989). Likewise, the macrosystem encompasses communal policies, cultural norms, and societal practices that influence both the microsystem and the exosystem (Bronfenbrenner, 1994). Finally, the chronosystem describes biological processes that shape a child's understanding, orientation, and belief system over time (Bronfenbrenner, 1994).

Within the context of school climate, the school itself functions as a microsystem, providing an environment for children's development (Rudasill et al., 2018). The formation of school climate involves collaboration among various actors, including teachers, students, and school leadership (Rudasill et al., 2018). Each member contributes unique personalities and experiences, which collectively shape the school climate. When different actors interact within the school, the mesosystem is actively engaged. Interactions between teachers and families (mesosystem) also have the potential to enhance children's learning experiences. The exosystem and mesosystem work together to develop cohesive school policies, which students are expected to follow within the school environment (Rudasill et al., 2018). As children grow within the school setting, the chronosystem may become evident through the diverse characteristics they exhibit. For example, students may vary in terms of learning difficulties or advanced intellectual abilities. The development of an ecological niche within the school suggests the need to accommodate the diverse learning needs of all students (Rudasill et al., 2018).

Building on EST, Rudasill et al. (2018) developed the Systems View of School Climate conceptual framework, which is situated within the microsystem. This framework explains various factors that influence child development within the school climate. Three components of school climate are identified: (a) social interaction and relationships, (b) a sense of physical and emotional safety, and (c) values and beliefs (Fig. 1).

In this study, these models are conceptualized as follows. First, social interaction and relationships refer to the degree to which students experience a supportive and conducive classroom environment. This is reflected in positive relationships with teachers and the development of a collaborative peer culture. A well-established peer culture is indicative of effective cooperation between students and teachers in classrooms. Second, the sense of physical and emotional safety represents the quality of relationships among students, fostering a sense of belonging and communication within an environment that prioritizes acceptance and inclusivity for all students. Third, values and beliefs refer to students' experiences and perceptions regarding academic disciplines such as science. These values may be shaped before students enter the microsystem, but understanding them is crucial for exploring their impact on child development.

In this study, the relationship between the three models (i.e., social interaction and relationships, sense of physical and emotional safety, and values and beliefs) and their impact on science achievement are assessed.

The relationship between school climate and science achievements

Research on the learning environment has consistently shown that

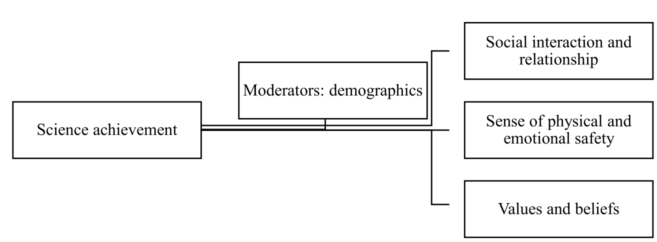

Fig. 1. Systems View of School Climate conceptual framework.

school climate impacts academic outcomes (Al-Qahtani, 2015; Rogers & Fraser, 2023), motivation levels (Alakoski, Laine & Hannula, 2025), and overall learning effectiveness (Al-Qahtani, 2015). Further, school climate is considered a vital component of providing high-quality education (Alakoski et al., 2025; Deed & Lesko, 2015). As a result, extensive research has been conducted to evaluate students' perceptions of their educational environment, the relationship between school environments and students' learning achievements, and some have even developed psycho-environmental model and tools to gauge the extent to which school climate may influence students' achievements. One example of the widely used tool is the Dundee Ready Education Environment Measure (DREEM) (Roff et al., 1997), which has been applied in various educational contexts (Arzuman, Yusoff & Chit, 2010; Jakobsoon, Danielsen & Edgren, 2011; Rogers & Fraser, 2023) and has demonstrated high reliability.

School climate, often considered a proxy for environmental factors, has been shown to have a significant compensatory effect on students' academic performance. However, given the variability in school environments and levels of teacher–student interaction, some scholars have found that the support perceived by students from teachers and peers is positively related to academic achievement but not necessarily to other developmental outcomes (Gómez & Suárez, 2020; Guzey & Li, 2023). In Qatar, findings from the Program for International Student Assessment (PISA) suggest that students' exposure to active methodologies, including practical and experimental work, may be generally engaging and motivating but less effective than anticipated in terms of academic achievement if implemented in an unsupportive school climate (Areepattamannil, 2014). In the UAE, Wardat et al. (2022) examined the impact of school climate on student achievement using the TIMSS 2015 database; they found a significant linear relationship between TIMSS science achievement and factors such as school resources, school discipline and safety, parental support, principal experience and education, and library and instructional resources.

Some researchers argue that school climate positively impacts student achievement (Benbenishty et al., 2016; Cheema & Kitsantas, 2014; Sakız, 2017) and is essential for academic success (Jones & Shindler, 2016). However, other studies have found no significant relationship or even a negative one between school climate and academic performance (e.g. Bear et al., 2014). These contradictory findings are often characteristic of situations where other intervening variables influence the relationship between the concepts (MacKinnon et al., 2002). Although among educational scholars there is no consensus on which aspects of school climate are the most important, broadly, there is an agreement that four interconnected factors—safety, teaching and learning, relationships, and the institutional environment—are crucial for science and math achievements (Alakoski et al., 2025; Arzuman et al., 2010; Bear et al., 2014; Rogers &Fraser, 2023).This necessitated our analyses of the 2019 dataset of the Trends in International Mathematics and Science Study (TIMSS) to explore the relationship between school climate and science achievement.

School climate as a mediating factor for science achievement in UAE: what do we know?

School climate is a multifaceted concept that encompasses safety, teaching quality, interpersonal relationships, and the overall institutional environment (Teng, 2020; Winberg & Palm, 2021). In the UAE, these elements are reflected in various initiatives developed to promote inclusivity, well-being, and academic excellence. For example, the National Child Protection Policy prioritizes physical and emotional well-being, ensuring respect for all students and inclusivity (MoE, 2021). UAE schools significantly emphasize fostering positive teacher–student interactions and peer support, which are crucial for cultivating a sense of belonging and inclusivity (Gningue et al., 2022). Furthermore, government initiatives such as the National programme for Happiness and Wellbeing Policy and the Dubai Inclusive Education Policy Framework reinforce the UAE's commitment to equity and inclusivity, particularly for students of determination (Ibrahim & Mahmoud, 2017; UAE Government, 2024). Programs such as the Moral Education initiative enhance the cultural relevance of education, integrating Emirati values while contributing to the development of positive school climates.

As the UAE transitions into a knowledge-based economy, fostering school environments that promote creativity and collaboration has become a strategic priority (AlKhaza'leh et al., 2023). Research consistently shows that a positive school climate not only improves academic outcomes but also supports students' socio–emotional development, aligning with the broader goals of the UAE's educational reforms (Thapa et al., 2013). For instance, Vision 2021 reflects the UAE's commitment to enhancing the quality of education to achieve global excellence. Attaining this vision requires an understanding of the key factors that shape school climate (Kippels & Ridge, 2019; Thapa et al., 2013).

A positive school climate characterized by safety, inclusivity, and collaboration (AlKhaza'leh et al., 2023) is strongly associated with increased student engagement and academic success. Specifically, El Zaatari (2020) found that factors such as positive teacher–student relationships, peer interactions, safety, inclusive classroom practices, and parental involvement significantly enhance students' sense of belonging in schools. In contrast, frequent exams and aggressive behaviors negatively impact this sense of belonging. Additionally, Aldridge and Khalil (2019) emphasized that supportive cooperative learning environments are essential for fostering student achievement, particularly in science education.

Most studies in the UAE have highlighted leadership as a cornerstone in promoting environments conducive to learning and improving the school climate (Madden, 2019; El Zaatari, 2020). Studies have shown that leadership practices, communication styles (Ibrahim & Mahmoud, 2017), and parental involvement (Alhosani et al., 2018) are particularly influential in shaping the school climate. However, studies in the UAE have predominantly focused on the role of leadership, with other critical dimensions receiving limited attention. Moreover, the specific impact of school climate on science education in the UAE remains underexplored, particularly regarding the direct influence of factors such as emotional safety and teacher support on science achievement. Addressing this gap, the present study aims to provide valuable insights into how school climate affects students' science achievement, offering actionable recommendations for education stakeholders to cultivate effective and inclusive learning environments.

Previous analysis of timss data

There is a small body of literature which has drawn on the TIMSS data to understand the synergy between school environment, attitude towards science and student achievement. Using 2003 TIMSS data, Chen et al. (2012) studied the impact of family resources, school climate and student participation in learning on science achievement among grade 4 and 8 students. They used the Science Environment Questionnaire which was operationalized as the interaction between students and teachers as well as the Attitude towards Science Questionnaire to assess its impact on one of the science plausible achievement data. Among the grade 4 students the data showed a positive effect of school climate and participation in leader on science achievement. However, among the grade 8 students, though school climate had positive effect on science achievement, negative relationship was found between participation in learning and achievement in science. In a similar study which drew on TIMSS 1995 data, Papanastasiou (2002) explored the factors which impacting on grade eight students' attitude towards science. Four items on School and Emotional Safety Questionnaire, the effect of reinforcement, teaching and family backgrounds on attitudes towards science which was measured using a four-item Attitude towards Science questionnaire. They result suggested an influence of school environment, family background and teaching on students' attitudes towards science.

This seems to suggest that regardless of the education level, school environment have an effect on students' achievement in science.

To the best of our knowledge, studies using TIMSS data to understand the impact of school environment on science achievement is scarce. This lends support to develop a comprehensive understanding of school environment and its effect on science achievement.

Current study

This study aimed to explore the relationship between science achievement and school climate from the perspective of students in the UAE. To develop an in-depth understanding of this phenomenon, TIMSS data of fourth-grade students were used. The students who participated in TIMSS also completed a series of questionnaires addressing key factors that could impact their performance. The literature showed a hypothesized relationship between school climate and students' achievement (Al-Qahtani, 2015; Benbenishty et al., 2016; Cheema & Kitsantas, 2014; Rogers & Fraser, 2023; Sakız, 2017). However, such relationship is yet to be assessed in an Arab context. The Moreover, the influence of demographic variables on the relationship between school climate and students' achievement is unresearched in an Arab context. The current study aimed to extend the literature on school environment and science achievement by testing the following hypotheses:

Hypothesis I: The core components of school climate (social interaction and relationships, sense of physical and emotional safety, and values and beliefs) are correlated.
Hypothesis II: School climate (social interaction and relationships, sense of physical and emotional safety, and values and beliefs) predicts the science achievement of fourth-grade students.
Hypothesis III: Demographic variables moderate the relationship between school climate (social interaction and relationships, sense of physical and emotional safety, and values and beliefs) and science achievement.

Method

The participants in this study were fourth-grade students who took part in the TIMSS during the 2023 cycle. TIMSS is administered by the IEA and has been conducted every four years since 1995 The goal of the TIMSS is to provide reliable data on fourth- and eighth-grade students' achievement in mathematics and science. In 2023, TIMSS was administered in 64 countries, including the UAE. The UAE has participated in TIMSS since 2007.

Since early childhood education has been prioritized by the UAE government (Abu Dhabi Early Childhood Authority, 2020), the current study drew on data collected from grade four students who participated in TIMSS in 2023. All public schools and selected private schools participated in the TIMSS in 2023 (Alzaabi et al., 2024; Ali et al., 2024). While public schools are free to UAE nationals, public schools utilize a number of curricula such as American, British, and Indian (Abu Dhabi Department of Education & Knowledge, 2024; Alzaabi et al., 2024; Ali et al., 2024). The public schools provide educational services to 10 % of school age population whereas private schools serve estimated 90 % of the students' population (Alzaabi et al., 2024).

The data used for this study were downloaded from IEA website (https://www.iea.nl/data-tools/repository/timss). In the 2023 TIMSS cycle, 34, 217 grade four students in private and public schools participated in the test. Table 1 summarizes the demographic characteristics of students who completed the TIMSS questionnaire in the UAE. The TIMSS student questionnaire collected extensive data from students (TIMSS, 2023). For the purpose of this study, both demographic data and questionnaire responses were used in the analysis. The research team reviewed the TIMSS instrument and discussed its operationalization for this study. After multiple meetings, the operationalization based on theoretical framework was shared with experts from the UAE Ministry of Education to finalize the selection of relevant variables for inclusion in the study. Demographic variables included students' gender, language of the test (i.e., language spoken at home), and absenteeism (see Table 1).

Table 1
Demographics of study participants .

Category (n = 34, 217)	Frequency	Percentage
Gender (n = 34,217)		
Females	16,784	49 %
Males	17,433	51 %
Language of the test spoken at home (n = 33,574)		
Always	10,351	31 %
Almost always	5748	17 %
Sometimes	14,295	43 %
Never	3180	10 %
Number of books at home (n = 33,525)		
0–10 books	5642	17 %
11–25 books	9480	28 %
26–100 books	9658	29 %
101–200 books	4602	14 %
>200 books	4143	12 %
Absenteeism (n = 34,842)		
Once a week	6061	18 %
Once every two weeks	2707	8 %
Once a month	3723	11 %
Once every two months	3725	11 %
Never or almost never	17,067	51 %

The next component of the study was an 11-item Social and Emotional Safety questionnaire (TIMSS, 2023). This aligns with students' sense of physical and emotional safety domain of the theoretical framework. Students were asked to rate the extent to which they had experienced certain behaviors from peers, including being made fun of or called names, being excluded from games, having lies spread about them, having something stolen from them, or having their belongings intentionally damaged. Responses were rated on a four-point scale from 1 (at least once a week) to 4 (never). Composite mean score of >3 was interpreted as favourable social and emotional safety. The reliability of the scale yielded a score of 0.93.

The next component, the Science Classroom Environment questionnaire (TIMSS, 2023), which aligns with the social interaction and relationship tenet of the study framework. This instrument measures to assess social interaction and relationship between teachers and students. This instrument consists of nine items, with responses ranging from 1 (agree a lot) to 4 (disagree a lot). Sample items included "I know what my teacher expects me to do," "My teacher is easy to understand," and "My teacher is good at explaining science." All items were positively worded. A mean score of at most 2 was interpreted as positive classroom environment. The reliability of the scale computed using Cronbach alpha yielded a score of 0.94.

The next component was operationalized as the Attitude Toward Science questionnaire (TIMSS, 2023). This instrument aligns with the values and beliefs tenet of the study framework. The instrument consisted of nine items rated on a four-point scale from 1 (agree a lot) to 4 (disagree a lot). Two items were negatively worded, whereas seven were positively worded. Sample items included "I enjoy learning science," "I wish I didn't have to study science," "Science is boring," "I like science," and "I look forward to learning science in school." A mean score of at most 2 was interpreted as positive attitude towards science. The reliability of the scale yielded a score of 0.72.

The fifth component, the Science Performance/Achievement questionnaire (TIMSS, 2023), was based on TIMSS measures of science achievement, which use five plausible data points. The total of the five plausible data were used for this study (Fishbein et al., 2023).

IBM Statistical Package for Social Science version 29 was used for data analysis. Before the analysis, Missing Completely at Random Test showed a non-significant result with missing data of <10 %. The Expectation maximization algorithm was used to replace the missing data.

Mean scores were computed to explore the level of school climate. Following this, the research questions were addressed. For Hypothesis I, Pearson-Moment Correlation Coefficient (Pallant, 2020) was used to examine the relationship between attitudes toward science, science classroom environment, and students' sense of safety (Pallant, 2020). The correlations were interpreted based on the following criteria: small (0.10–.29), medium (0.30–.50), and large (at least 0.51; Pallant, 2020).

For Hypothesis II, hierarchical multiple regression was computed to explore the relationship between school climate and science achievement. The domains of the framework (social interaction and relationships [Science Classroom Environment], sense of physical and emotional safety [Social and Emotional Safety], and values and beliefs [Attitude Toward Science]) were used as independent variables and regressed directly on science achievement in Step 1. Following this, in Step 2, the demographic variables (gender, number of books at home, language of the test spoken at home, and absenteeism) were added to the model to explore its unique contribution in the variance in science achievement. Assumptions of homoscedasticity, linearity and multicollinearity were checked to ensure that they were not violated (Pallant, 2020).

For Hypothesis III, Andrew Hayes' (2022) Moderation Analysis Method 1 was used to explore the effect of the demographic variables on the relationship between school climate and science achievement. Each of the demographic variables were used as moderators to explore its effect on the relationship between tenets of climate change (independent variable) and science achievement (dependent variable). Moderation effect as assumed in the event where there is a significant relationship between dependent and independent variables as well as moderator having a direct significant effect on the former. The alpha value was set at 0.05 (95 % confidence interval).

Result

Initial computation of mean scores showed as follows: Attitude Toward Science ($M = 1.83$, SD = 0.49), Science Classroom Environment ($M = 2.59$, SD = 1.02) and Social and Emotional Safety ($M = 3.10$, SD = 0.86).

Hypothesis I: relationship between domains of science environment

Table 2 summarizes the correlation between the variables. For instance, large correlation was found between attitude toward science and science classroom ($r = 0.52$, $p = .001$) whereas moderate correlation was found between science achievement and social and emotional safety ($r = 0.36$, $p = .001$). Small correlations were found between the other variables. see Table 2 for more details.

Hypothesis II: influence of science environment on achievement

The predictors of science achievement were explored (see Table 3). In step 1, the independent variables (Attitude Toward Science, Science Classroom Environment and Social and Emotional Safety) were regressed directly on science achievement. The three predictors made 12 % contributions in the variance in science achievement, $R^2 = 0.12$; F (3, 28,253) = 13,114.64, $p = .001$. Here, all the three independent variables contributed significantly in the variance in science achievement.

In step 2, the demographic variables were added to the model which made 6 % contribution in the variance in science achievement, R2 change = 0.06; F (4, 28,249) = 471.72. The combined demographics and independent variables made 18 % contributions in the variance in science achievement, R2 = 0.18; F (7, 28,249) = 870.52, $p = .001$. Although all the demographics and independent variables made individual contributions in the variance in science achievement, social and emotional safety made more unique contributions.

Table 2
Correlation between science classroom environment and achievement .

		1	2	3	4
1	Science Classroom Environment				
2	Attitude Toward Science	.52**			
3	Social and Emotional Safety	−0.17**	.10**		
4	Science Achievement	−0.21**	−0.10**	.36**	

Note: **$p \leq .01$.

Table 3
Predictors of science achievement .

	Stand. Beta	t	p	Confidence Interval	
				Lower	Upper
Step 1					
Science Classroom Environment	−0.17	−25.51	.001**	−36.25	−31.07
Attitude Toward Science	.02	3.15	.002**	1.70	7.33
Social and Emotional Safety	.28	49.31	.001**	33.01	35.75
Step 2					
Science Classroom Environment	−0.15	−23.16	.001**	−32.22	−27.19
Attitude Toward Science	.02	2.57	.01**	.85	6.32
Social and Emotional Safety	.25	44.68	.001**	29.40	32.10
Gender	.05	9.89	.001**	9.10	13.59
Language of the test	−0.04	−6.48	.001**	−4.86	−2.60
Number of books at home	.03	5.96	.001**	1.87	3.71
Absenteeism	.22	40.59	.001**	14.38	15.84

Note: **$p \leq .01$; stand. Beta = standardized Beta.

Hypothesis III: moderation analysis

Moderators of science classrooms environment and achievement

Number of books at home moderated (beta = −2.48, $p = .003$, 95 % CI [−4.12, −0.83]) the relationship between science classroom environment and achievement, R2 = 0.05; F (3, 28,932) = 478.44, $p = .001$ (see Table 4).

Individually, science classrooms environment and Number of books at home had direct impact on science achievement. In the event participants had 0–10 books (beta = −37.08, p = .001, 95 % CI [−40.76, −33.40]), 26–100 books (beta = −42.03, p = .001, 95 % CI [−44.27, −0.39.79) or 101–200 books at home (beta = −44.51, p = .001, 95 % CI [−47.49, −41.53]), there was a relationship between science classrooms environment and achievement. From Fig. 2, it was evident that when classroom environment was favourable, those having between 101–200

Table 4
Moderators of science classrooms environment and achievement.

	Beta	t	p	Confidence interval	
				Lower	Upper
Gender	−2.92	−0.90	.37	−9.30	3.45
Science Class Env.	−51.95	−13.91	.001**	−59.27	−44.63
Science Class Env. *X* Achievement	6.34	2.80	.01*	1.90	10.78
Language of the test	1.03	.66	.51	−2.03	4.08
Science Class Env.	−28.29	−9.69	.001**	−34.01	−22.57
Science Class Env. *X* Achievement	−4.85	−4.51	.001**	−6.95	−2.74
Number of books at home	7.42	5.93	.001**	4.96	9.87
Science Class Env.	−34.60	−13.32	.001**	−39.69	−29.51
Science Class Env. *X* Achievement	−2.48	−2.95	.003**	−4.12	−0.83
Absenteeism	12.93	13.08	.001**	10.99	14.87
Science Class Env.	−49.71	−19.05	.001**	−54.93	−44.60
Science Class Env. *X* Achievement	3.72	5.49	.001**	2.39	5.05

Note: **$p \leq .01$; * $p \leq .05$; stand. Beta = standardized Beta.

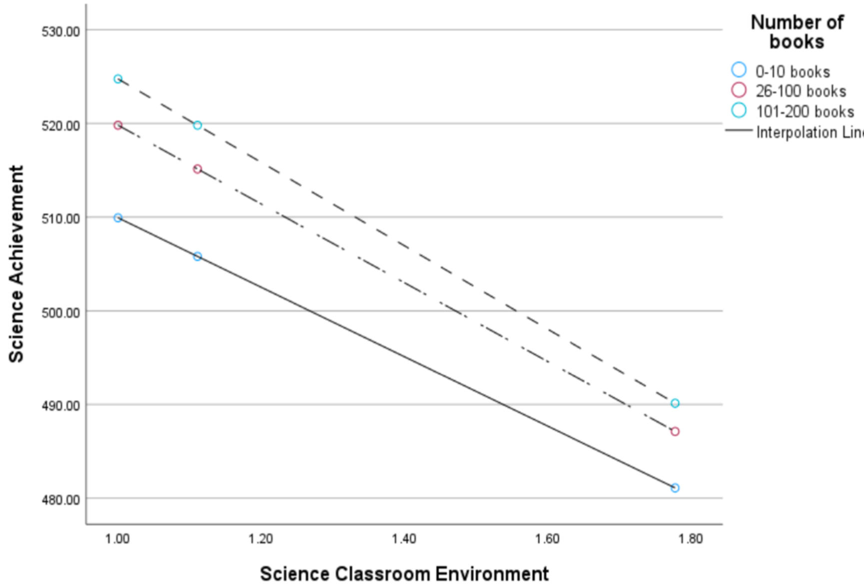

Fig. 2. Number of books as a moderator between science classroom environment and achievement.

books had higher performance compared to those having 0–10 books and 26–100 books. Similar trends were observed when science classroom environment decreases.

Absenteeism moderated (*beta* = 3.72, *p* = .001, *95% CI* [2.39, 5.05]) the relationship between science classroom environment and science achievement, R2 = .12; F (3, 28841) = 1255.21, p = .001. Individually, absenteeism and science classroom environment had direct impact on science achievement. When participants were absent once a week (beta = -45.99, p = .001, 95% CI [-49.94, -42.04]) or never (beta = -31.10, p = .001, 95% CI [-34.02, -28.17), significant relationship was found between classroom environment and science achievement. From Fig. 3, it could be seen that when science classroom environment is favourable, the performance of students who were never absent performed better than those who were absent once a week. Similarly, as science classroom environment declines, those who were never absent performed better than those who were absent once a week.

Moderators of attitude toward science and achievement

The moderators of the relationship between attitude towards science and achievement were explored (see Table 5). Only absenteeism significantly moderated (beta = 2.68, *p* = .001, 95 % CI [1.28, 4.08]) the relationship between attitude towards science and science achievement, R2 = 0.10; F (3, 33,279) = 1209.27, *p* = .001. Both absenteeism and attitude toward science had direct effect on science achievement. In the event students are absent once a week (beta= −25.76, *p* = .001, 95 % CI [−29.82, −21.72) or never, (beta = −15.05, *p* = .001, 95 % CI [−18.23, −11.87) significant relationship was found between attitude towards science and science achievement. From Fig. 4, it could be observed that when attitudes were favourable, students were never absent, their performance was higher than those who indicated that they were absent once week. Similar trend was observed when attitudes towards science appear to decrease.

Moderators of social and emotional safety and achievement

The moderators of the relationship between social and emotional

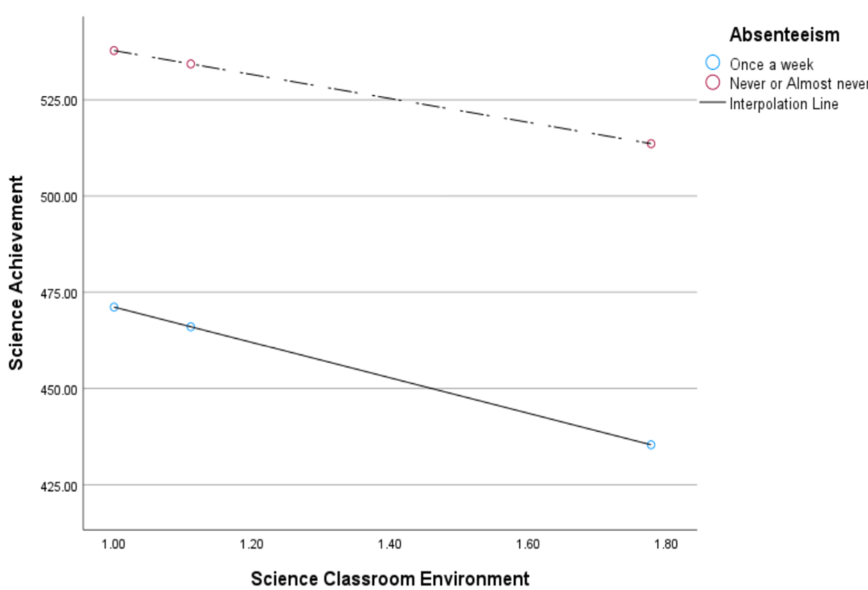

Fig. 3. Absenteeism as a moderator of science classroom environment and achievement.

Table 5
Moderators of Attitude Toward Science and achievement.

	Beta	t	p	Confidence interval	
				Lower	Upper
Gender	.34	.07	.94	−8.68	9.36
Attitude Toward Science	−20.78	−5.22	.001**	−28.59	−12.99
Att. Toward *Sc. X* Achievement	−0.17	−0.06	.94	−4.92	4.59
Language of the test	9.91	4.64	.001**	5.73	14.09
Attitude Toward Science	3.49	1.19	.23	−2.24	9.22
Att. Toward *Sc. X* Achievement	−9.46	−8.44	.001**	−11.66	−7.27
Number of books at home	−0.27	−0.15	.88	−3.68	3.15
Attitude Toward Science	−24.58	−8.79	.001**	−30.04	−19.08
Att. Toward *Sc. X* Achievement	1.44	1.59	.11	−0.33	3.22
Absenteeism	15.88	11.80	.001**	13.24	18.52
Attitude Toward Science	−28.45	−10.59	.001**	−33.71	−23.18
Att. Toward *Sc. X* Achievement	2.68	3.76	.001**	1.28	4.08

Note: $^{**}p \leq .01$; $^{*}p \leq .05$; stand. Beta = standardized Beta.

safety and science achievement were assessed (see Table 6).

First, gender moderated (beta = −3.71, p = .004, 95 % CI [−6.26, −1.16]) the relationship between Social and Emotional Safety and achievement, R2 = 0.11; F (3, 34,213) = 1407.52, p = .001. Gender and social and emotional safety made individual contribution in the variance in science achievement. In the event participants were females (beta = 44.11, p = .001, 95 % CI [42.23, 45.98]) or males (beta = 40.40, p = .001, 95 % CI [38.67, 42.13]), significant relationship were found between Social and Emotional Safety and achievement. From Fig. 5, whether social and emotional safety were high or low, males achieved higher than females.

Second, language of test moderated (beta = 2.55, p = .001, 95 % CI [1.33, 3.77]) the relationship between Social and Emotional Safety and achievement, R2 = 0.11; F (3, 33,570) = 1401.40, p = .001. language of test and social and emotional safety had direct significant impact on science achievement. Whether the participants speak the language of the test always (beta = 37.71, p= .001, 95 % CI [35.72, 39.70]) or sometimes (beta = 42.81, p = .001, 95 % CI [41.24, 44.38]), there was a significant relationship between Social and Emotional Safety and achievement. From Fig. 6, whether social and emotional safety is either low or high, students who always speak the language of the test at home performed better than those who sometimes speak the language of the test.

Third, number of books moderated (beta = 3.54, p = .001, 95 % CI [2.55, 4.53]) the relationship between Social and Emotional Safety and achievement, R2 = 0.11; F (3, 33,521) = 1385.38, p = .001. In the event participants indicated they have 0–10 books (beta = 35.17, p = .001, 95 % CI [32.98, 37.36]), 26–100 books (beta = 42.25, p = .001, 95 % CI [40.96) or 101–200 books (beta = 45.79, p = .001, 95 % CI [44.05, 47.53]), significant relationship was found between Social and Emotional Safety and achievement. From Fig. 7, it could be observed that when Social and Emotional Safety is low, no difference in achievement was found between students based on number of books at home. However, as social and emotional safety increases, those having 101–200 books performed higher than those having 0–10 books and 26–100 books at home.

Additionally, absenteeism moderated (beta = −0.19, p = .02, 95 % CI [−1.65, −0.13]) the relationship between Social and Emotional Safety and achievement, R2 = 0.17; F (3, 33,279) = 2263.55, p = .001. Absenteeism and social and emotional safety had direct impact on science achievement. Either participants were absent once a week (beta = 38.12, p = .001, 95 % CI [35.83, 40.40]) or never (beta = 34.56, p = .001, 95 % CI [32.88, 36.25]), significant relationship was found between Social and Emotional Safety and achievement. From Fig. 8, it was observed that the performance of students who were never absent performed better than those who were absent once a week. As social and emotional safety increases, students' performance also appreciates.

Table 6
Moderators of Social and Emotional Safety and achievement.

	Beta	t	p	Confidence interval	
				Lower	Upper
Gender	19.22	4.57	.001**	10.98	27.45
Social and Emotional Safety	47.82	22.71	.001**	43.69	51.94
Social and Emotional Safety *X* Achievement	−3.71	−2.85	.004*	−6.26	−1.16
Language of the test	−13.82	−6.88	.001**	−17.75	−9.88
Social and Emotional Safety	35.16	22.73	.001**	32.13	38.19
Social and Emotional Safety *X* Achievement	2.55	4.09	.001**	1.33	3.77
Number of books at home	−8.00	−5.02	.001**	−11.12	−4.87
Social and Emotional Safety	31.64	20.35	.001**	28.59	34.68
Social and Emotional Safety *X* Achievement	3.54	7.02	.001**	2.55	4.53
Absenteeism	20.06	16.36	.001**	17.66	22.46
Social and Emotional Safety	39.01	25.91	.001**	36.06	41.96
Social and Emotional Safety *X* Achievement	−0.89	−2.29	.02*	−1.65	−0.13

Note: $^{**}p \leq .01$; $^{*}p \leq .05$; stand. Beta = standardized Beta.

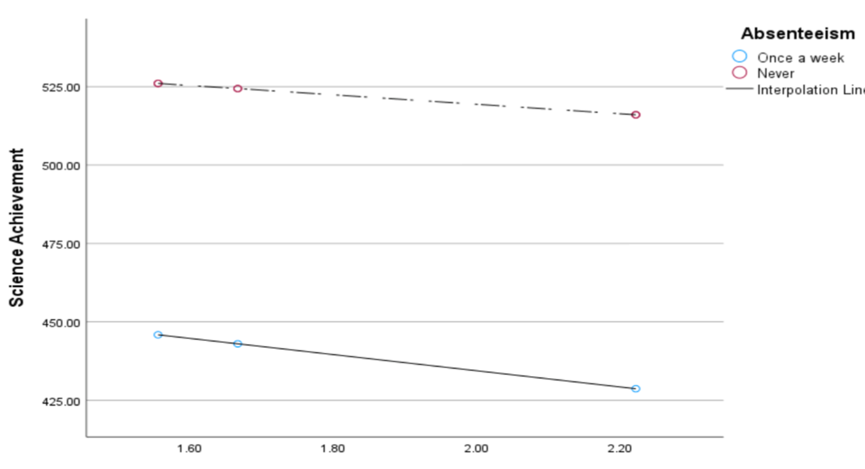

Fig. 4. Absenteeism as a moderator of Attitude Toward Science and achievement.

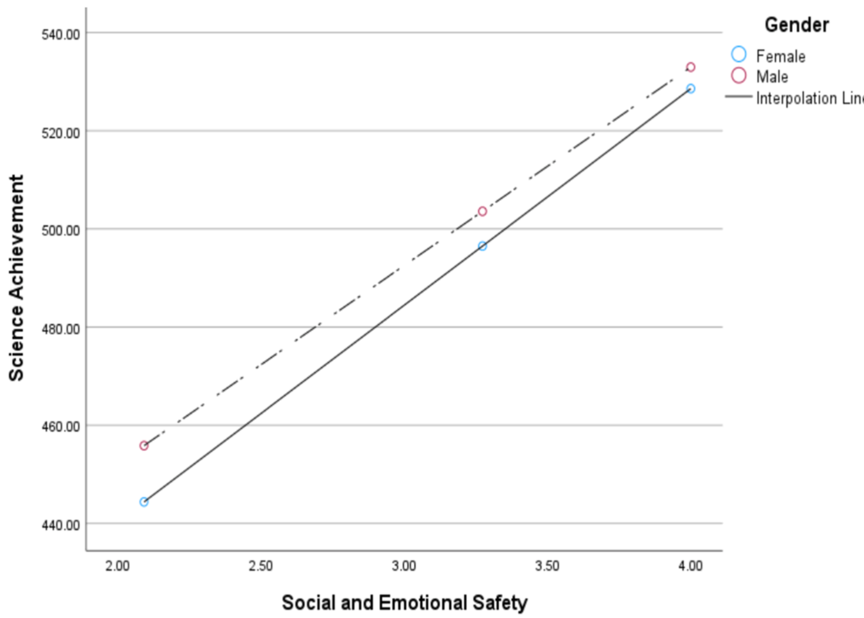

Fig. 5. Gender as a moderator of Social and Emotional Safety and achievement.

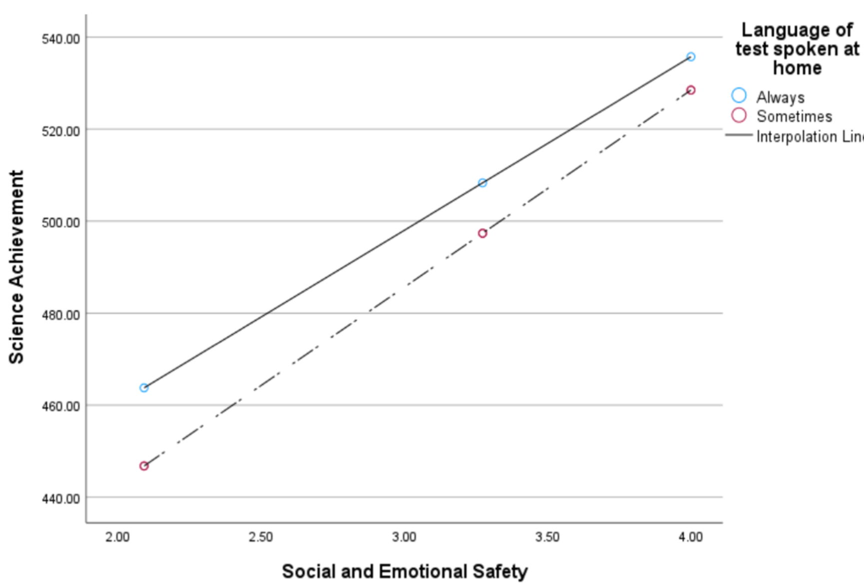

Fig. 6. Language of test as a moderator of Social and Emotional Safety and achievement.

Discussion

The current study extends research on learning environment from western context to the UAE where there is scarcity. Drawing on grade four students' TIMSS data, the study has added to the literature the synergy between school climate (used as a proxy for learning environment) and students' achievement in science. The current study is relevant and contributes to contemporary discussions on learning environment. Threefold hypotheses guided the study; while two were fully supported (Hypotheses I and II), one was partially supported (Hypotheses III).

The study findings provided support for Hypotheses I; association were found between the tenets of school climate. This finding supports Rudasill et al.'s (2018) Systems View of School Climate conceptual framework which suggested synergy between social interaction and relationship, physical and emotional safety as well as values and beliefs. According to Rusticus et al. (2020, 2023), school climate encompasses school related factors such as physical, emotional and psychological which could have impact on the learning of students. The UAE is attempting to improve students' achievement in science. Previous studies have shown that there are notable barriers such as teachers, resources and lack of students' motivation (Teng, 2020; Wardat et al., 2022) which is adversely impacting on students' achievement in science. These barriers identified in the UAE align with Rudasill et al. (2018) conception of school climate. The nature of the school cannot be dissociated from such discussion. Indeed, the study has shown that multifaceted effort may be needed to advance the teaching of science in the UAE context.

In this study, interesting relationships were found between tenets. For instance, large positive correlation was found between social interaction and beliefs of students. Although it is not our intention to assume causation, it could be inferred that developing social interaction

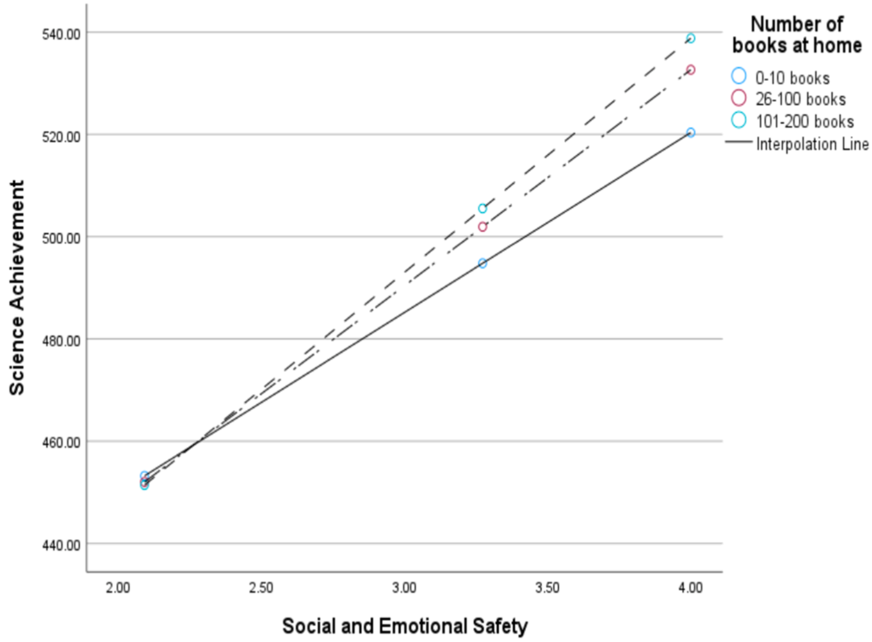

Fig. 7. Number of books as a moderator of Social and Emotional Safety and achievement.

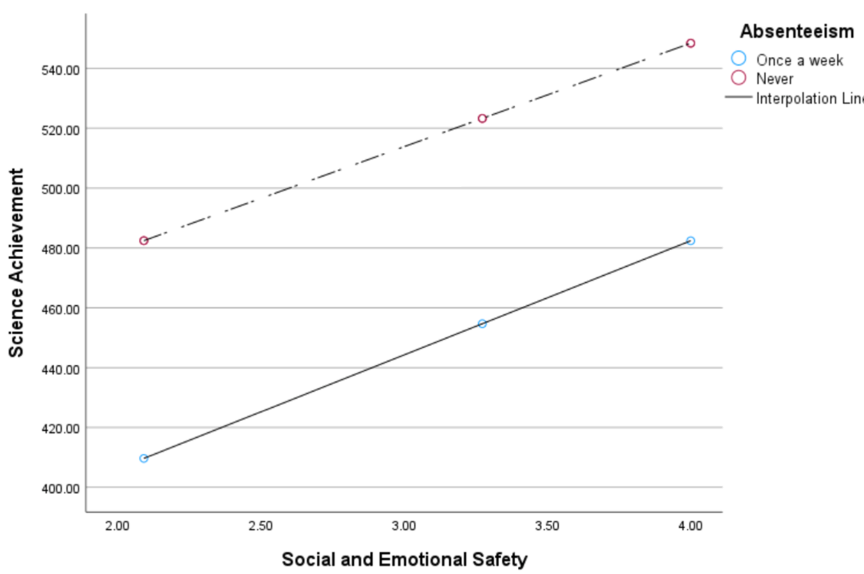

Fig. 8. Absenteeism as a moderator of Social and Emotional Safety and achievement.

between students could affect students' beliefs towards science. Improving teacher-student interaction in science classroom could positively impact on their attitude towards science. The finding reported here could be linked to the centrality of teachers to promoting students' learning (Berkowitz et al., 2015; Guzey & Li, 2023; Winberg & Palm, 2021). Consistently, teachers have been identified as the mainstay in education (Badri, 2019; Guzey & Li, 2023; Teng, 2020). The study has shown that teaching science teachers to probably maintain healthy relationships in classrooms could improve students' attitudes towards science. Attitude is perceived as an important variable which indicates whether an individual could engage in a given activity or not (Erdem & Kaya, 2024; Teng, 2020; Wardat et al., 2022). In contemporary classroom, healthy relationships between students and teachers is very crucial for successful teaching practices (Erdem & Kaya, 2024; Gómez & Suárez, 2020). Without deliberate and cordial relationship between students and teachers in science classrooms, there is potential for the former to develop unfavourable attitude towards science (Badri, 2019; Jiang, Chen & Wu, 2021; Wardat et al., 2022). The study has shown that in the event teacher educators are working towards improving students' attitude towards science, teaching teachers to maintain healthy relationships with students could be part of teacher training and professional development training.

Hypothesis II was also supported with domains of school climate combining to predict science achievement. Each of the three domains made significant contribution in the variance in science achievement. Existing research has established a clear connection between school climate and science achievement, with findings demonstrating that regular school attendance is directly associated with improved learning outcomes (Hammer, Melhuish & Howard, 2018; Vesić, Džinović & Mirkov, 2021; Wardat et al., 2022; Zhang & Bae, 2020). However, there were disparities in the impact of school climate on science achievement. In this study, sense of physical and emotional safety made more significant contribution in the variance in science achievement. It could be inferred that for every unit of increase in sense of physical and emotional safety, students' achievement in science increases by 28 %. For decades, empirical studies have consistently found strong correlations between social and emotional safety, attitude toward science, and science achievement (Badri, 2019; Cristóvão, Candeias & Verdasca, 2017; Wardat et al., 2022). Longitudinal evidence further suggests that socio-emotional behaviors, including emotional safety and student attitudes, influence science learning outcomes (Hammer et al., 2017; Hammer, Melhuish & Howard, 2018). Under Maslow hierarchy of needs, safety of individual is next to basic individual needs (Gambrel & Cianci, 2003). Without safety, students may feel unsettled to participate in learning activities. In order to improve students' achievement in science in the UAE, creating a sense of safety seems to be a priority for students who took part in this study.

Social interaction and relationships also made enormous contributions in the variance in science achievement. Specifically, it was found that for every increase in student-teacher interaction, science achievement could improve by 17 %. Research has consistently documented that the quality and stability of interpersonal relationships in schools significantly influence children's development (Demirtas-Zorbaz et al., 2021; Fan & Williams, 2018). Consequently, school climate has the potential to affect multiple dimensions of student growth. Studies that have examined how school-related factors, such as the role of school personnel in fostering a supportive environment, shape student outcomes suggest that socio-emotional skills and supportive environments enhance students' potential for science achievement (Fan & Williams, 2018; Hammer, Melhuish & Howard, 2018; Ndijuye & Benguye, 2023). It could be surmised that when students feel safe, they are more likely to explore their potential without fear. Also, they are more likely to participate and excel in science. These findings indicate that policymakers and educators should prioritize developing students' attitudes toward science and fostering safe learning environments for all students.

Hypothesis III was partially supported as differences were found in the moderating effect of demographics on the relationship between school climate and science achievement. For example, absenteeism had a moderating effect on the relationship between the three domains of social climate and science achievement. These findings align with those of Vesić and colleagues (2021), who analyzed TIMSS 2015 data to examine the moderating role of absenteeism in science achievement in Serbia. In this study, the findings showed that students who were never absent performed better on school climate and science achievement compared to those who were absent once a week. This finding is tenable in the sense that when students are absent, they are unable to follow teaching and learning activities (Oberleiter et al., 2023). Although it was beyond the scope of this study to gather information relating to factors contributing to absenteeism, hostile school environment in the form of peer abuse could impact on school attendance (Oberleiter et al., 2023; Winberg & Palm, 2021; Zhang & Bae, 2020). It is essential for educators to be interested in factors which account for student absenteeism in the UAE. This has the potential to affect interventions which could be institutionalized by educators to promote attendance and science achievement.

Moreover, the number of books a student has at home moderated the relationship between two tenets of school climate and science achievement. Specifically, the more the number of books a student has at home, the higher students' relationship with teachers, sense of safety and higher the achievement in science (Prinsloo, Rogers & Harvey, 2018; Zhang & Bae, 2020). In previous studies, home learning support has been found to impact positively on students' achievement (Hammer, Melhuish & Howard, 2018; Oberleiter et al., 2023; Vesić, Džinović & Mirkov, 2021; Wardat et al., 2022; Zhang & Bae, 2020). In this study, it appears quantity of books of students have at home could affect their achievement in science. There is interrelationship between an environment a student is raised in and their development (Fan & Williams, 2018; Wardat et al., 2022; Winberg & Palm, 2021;). It appears making reading materials available to students has the potential to impact on students' interactions with teachers and emotional development. Educators could include parents in effort towards promoting students' achievement in science. This could be in the form of training parents to understand the role of home-learning support in enhancing students' achievement in science (Oberleiter et al., 2023).

Study limitations

This study examined the relationship between school climate and science achievement using the 2023 TIMSS dataset in the UAE. As the study focused exclusively on TIMSS data, we did not compare our findings with data from other international assessments such as PISA. Additionally, although we acknowledge that science achievement can be influenced by multiple factors, our study was limited to selected variables (i.e., gender, language, and absenteeism). Furthermore, our findings may not fully reflect the broader implications of these variables across the UAE school system because the analysis was restricted to fourth-grade students. We recommend that future studies extend the scope to include multiple grade levels across the country to provide a more comprehensive understanding of these relationships. Similarly, future research could analyze and compare datasets from other Gulf countries that participate in TIMSS assessments. Such comparative analyses may offer a more holistic perspective on the relationship between school climate and science achievement in the Middle East and North Africa region.

This study did not explore other dimensions of school climate, such as school leadership and parent–school partnerships. School leadership plays a critical role in driving educational change, overseeing teacher development, fostering parental engagement, maintaining a positive school climate, and providing instructional leadership (Hammer et al., 2017). Future research could further investigate the role of school leadership in shaping students' achievement in science.

Conclusion and implication for practice

This study aimed to explore the relationship between science achievement and school climate using the data of fourth-grade UAE students in the 2023 TIMSS dataset. This study was conducted against the backdrop of increasing recognition of school climate as a determinant of student achievement and its limited exploration within the context of science learning in an Arab setting such as the UAE. Findings indicated that social interaction, emotional safety and students' attitudes toward science had a direct effect on science achievement. Additionally, demographic factors such as language of the test and absenteeism exhibited both direct and indirect effects on science achievement. These findings provide valuable insights for policymakers seeking to promote science education in the UAE.

The UAE government has introduced numerous initiatives aimed at enhancing science education. However, our findings suggest that tenets of school climate such as social interaction, safety and beliefs are key predictors of science achievement. Educators could engage school stakeholders in ways they could create supportive learning environment for students in early years of education. National dialogue could be conducted nationwide to discuss the importance of school environment and supportive measures in schools and classrooms. Additionally, educators could consider designing professional development for teachers addressing each of the tenets of school climate as they could promote science achievement. This has the potential to equip teachers with practical skills to create conducive science learning environment for all students.

Furthermore, schools could implement positive behavior intervention programs, which have shown to foster a conducive learning environment. Awareness campaigns, such as posting positive and welcoming messages through signage and training school personnel to address negative student behaviors, could further enhance school climate.

Moreover, absenteeism was found to be a significant moderator of the relationship between social and emotional safety and science achievement. Addressing absenteeism could therefore improve student performance. Schools could enhance engagement between school leaders, students, and parents to better understand the root causes of absenteeism. By identifying and addressing these factors, educators could help increase student participation and improve science achievement. Furthermore, parents could be trained in their contribution to the education of their children. This could be in the form of organizing parental meetings, workshops or symposium on best reading practice such as making books available to children at home. Without adequate literacy skills, students may struggle to perform well in science.

This study is novel as it relies on international datasets to provide insights into the relationship between school climate and science achievement from the perspective of fourth-grade students. Additionally, it contributes to the literature by identifying factors that can positively influence science achievement in an Arab context such as the UAE.

Statements and declarations

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Ethics approval

N/A

Availability of data and materials

The data is available on request from the corresponding author.

CRediT authorship contribution statement

Hassan Tairab: Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Funding acquisition, Data curation, Conceptualization. **Latifa Alketbi:** Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Conceptualization. **Ahmad Qablan:** Writing – review & editing, Writing – original draft, Resources, Methodology, Conceptualization. **Laurent Gabriel Ndijuye:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Maxwell Peprah Opoku:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article

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